Group 53 Progress Report:

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1 Introduction

The Swing Trading Strategy Analyzer tests if machine learning can help decide whether a setup is strong or weak for the moving-average pullback strategy specifically. A rule-based detector scans 4-hour bars across 76 stocks. It flags a setup when a stock is trending above the 50-bar simple moving average (SMA), then pulls back into its 20-bar exponential moving average (EMA), and then continues moving up.

The machine learning model gives each setup a probability of being successful. Every trade has a stop loss $1.5\times$ the average true range (ATR) under the entry and a profit target $3\times$ the ATR above it, with a 10-bar time limit. A success means the target is reached before the stop loss is hit and before the time limit is over. The model uses price action features and macro features influenced by the overall market. Features include news tone from GDELT and the VIX volatility index.

This problem is difficult for 3 reasons:

1. There is a lot of noise in the market, so models can be wrong more often than not.
2. Financial machine learning results can be inflated by leakage, where a feature accidentally uses information from after the trade decision.
3. News data comes from a different source than the price data, so it has to be lined up carefully in time.

Due to this, we are not aiming for a super accurate model. We are aiming for one that is correct and does not leak. The project is a success if filtering setups with the model gives better trading results than taking every setup the detector finds or than just buying and holding.

2 Related Work

Rhea will add her part today!!!

3 Dataset

You should write about your dataset here, following the guidelines regarding item 1. This section may be 0.5-1 pages. Depending on your specific dataset, you may want to include subsections for the preprocessing, annotation, etc.

4 Features

Each setup is one row of data with 21 features. They go straight into the tree models, with no embeddings or neural network. Stock prices vary a lot, so every price-based turns into a percentage or ratio. This is so all tickers are on the same scale.

Detector features:

- RSI
- ATR is percent of price
- how far the close price is above EMA20
- how far the close price is above SMA50
- trend strength is how far the EMA20 is above the SMA50
- if the signal came on the morning or afternoon bar

Technical features: add information the detector doesn't use:

- MACD line and histogram divided by the close price
- change in the MACD histogram over the 3 pullback bars, where positive means momentum is turning upward
- 20-bar volatility of log returns
- 5-bar vs. 20-bar volatility ratio, and above 1 means volatility is getting more intense

- signal bar volume compared to the average of the last 20 bars from the same session (the morning bar is 4 hours and the afternoon bar is 2.5 hours, so they can't be mixed)
- average relative volume during the pullback where low means a quiet dip and not heavy selling

Macro features: overall market tone when the signal fires:

- VIX: latest close, its 5-day change, and its distance from its 20-day average
- GDELT, from 72 hours before the signal bar closes:
  - average news tone
  - average Goldstein score (a scale from conflict to cooperation)
  - average tone of U.S. events
  - change in tone (last 24 hours vs. the last 48 hours)
  - news volume ratio (last 24 hours vs. the last 48 hours)

Leakage prevention:

- Each feature table saves the timestamp of the newest data it used, and a check makes sure none of them are later than the signal bar's close.
- The VIX daily close only counts as available from the next day.
- A GDELT hour only counts once it is fully over.
- We also run a truncation test. It cuts the raw data off right after random setups, rebuilds the features, and checks that values for earlier setups do not change. If they did, the feature would be using future data.
- Separate checks were run for the technical and macro features.

Feature selection. We also tested relative strength features against SPY: a 20-bar relative return, a short-term relative return, and a rank across tickers. They didn't improve the walk-forward results, so we left them out. The effect of each feature group is shown in the ablation in the Implementation section.

5 Implementation

Describe your model and implementation here. Refer to item 4. This may take around a page.

6 Results and Evaluation

Splitting: Setups are sorted by signal time. The last 20% (268 setups) is the final test set and the first 80% (1,038 setups) is used for training. A random split would let the model learn from setups that happen at the same time as the test setups, which inflates results. We also use a 3-day embargo, so training trades have to close at least 3 days before the test window starts. This means trade outcomes do not overlap with the test period.

Cross-validation: Hyperparameters are only tuned on the training set, they use expanding-window walk-forward folds.

- The training data is split into 5 chunks in time order.
- Fold k trains on chunks 1 to k and validates on chunk $k + 1$, with the same embargo.
- Settings are picked by the average PR-AUC across the folds.

To have a stronger check, a full walk-forward run retrains the model across 8 time windows over the whole history and scores each window on data it has not seen.

Classification metrics: Only about 21% of test setups are wins, which makes it inaccurate. These get reported:

- precision, which is the amount of predicted wins that actually win
- recall, which is the share of actual wins that the model predicts as wins
- PR-AUC, compared against the 21% base rate
- ROC-AUC, 0.5 means random ranking
- TSS (true skill statistic), which is recall minus the false positive rate, where 0 means no skill

Trading metrics: Trades measured in R, which is profit in units of the risk taken. A full win is $+2$R and a full stop out is -1R. We report win rate, average R per trade, Sharpe ratio and max drawdown. To compare with buy-and-hold percentage returns, each trade is assumed to risk 1% of the account.

Test set (268 setups)	XGBoost	LightGBM
ROC-AUC	0.49	0.46
PR-AUC	0.21	0.20
Precision	21.1%	18.8%
Recall	26.8%	26.8%
TSS	0.004	−0.04

Table 1: Classification result for the final test set. The base rate is 20.9% wins.

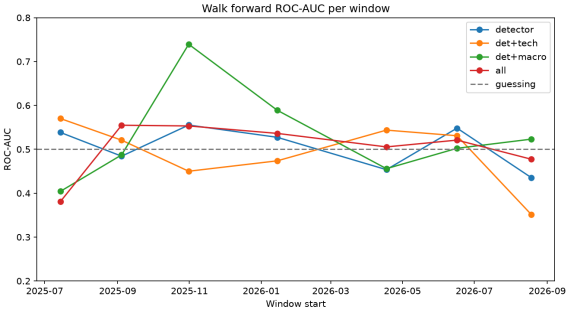


Figure 1: Walk-forward ROC-AUC for each time window and feature set, the values stay close to 0.5.

Baselines:

- buy and hold SPY and QQQ
- equal weight buy and hold of all 76 tickers
- the unfiltered detector takes every setup

The unfiltered detector is the main baseline. If the model does not beat it, the model does not add value.

Results so far: Table 1 shows the test set results. Neither model decides winners against losers better than chance, and both have precision almost equal to the base rate.

While tuning, the best settings scored about 0.30 PR-AUC, compared to fold base rates of 0.20 to 0.27. This looked like a small edge, but it disappeared on the test set. Picking the best of 24 (XGBoost) or 36 (LightGBM) settings on only about 1,000 rows gives a score that is good but only because of luck. Both models also picked the largest minimum leaf size, which is what tuning tends to do when there not much of a real pattern to find.

The walk-forward results are similar (Figure 1). Across both models and all the feature sets, ROC-AUC is between 0.49 and 0.53, and every 95% confidence interval has 0.5. ROC-AUC for individual windows ranges from 0.35 to 0.74 where there s no steady trend, which indicates luck and not a skill that will hold over time.

Table 2 shows the backtest over the approximately 90-day test window.

	Return	Sharpe	Max DD	Trades
SPY hold	+3.5%	0.91	3.5%	–
QQQ hold	−4.5%	−0.53	10.3%	–
Equal-wt hold	+9.9%	2.11	3.3%	–
Unfiltered detector	+4.1%	0.47	18.8%	268
XGBoost ($p \geq 0.5$)	+2.2%	0.43	7.8%	71

Table 2: Backtest over the final test window. Each trade risks 1% of the account. The equal weight hold is rebalanced every day.

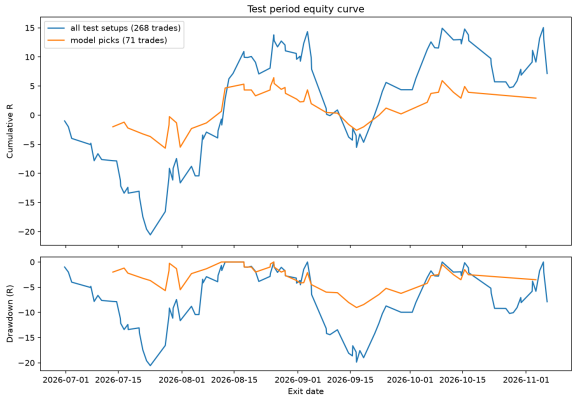


Figure 2: Cumulative R for model filtered trades compared to all of the detector setups.

- The XGBoost filter took 71 of the 268 setups.
- It returned +2.2% (Sharpe 0.43), compared to +4.1% (Sharpe 0.47) for the unfiltered detector.
- It reduced the max drawdown from 18.8% to 7.8%, mostly because it took fewer trades.
- The equal weight buy and hold returned +9.9% (Sharpe 2.11), which is higher than both.

The model's picks earned a little more per trade than all the setups on the test set, 0.041R compared to 0.027R (Figure 2). With ROC-AUC around 0.5, it is likely that the difference is due to noise. Total R was also lower for the model (2.9R vs 7.2R) because it took less trades.

Over the full history, the unfiltered detector averaged +0.157R per trade over 1,315 trades. This suggests the pullback pattern is mildly positive on its own, but our current features do not predict which setups will work.

7 Feedback and Plans

Write about your plans for the remainder of the project. This should include a discussion of the feedback you received from your TA, and how you

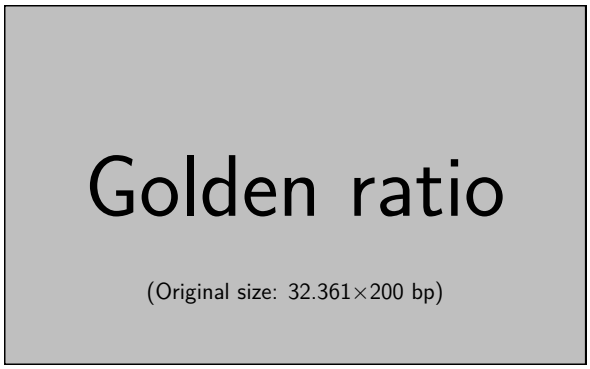


Figure 3: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

plan to improve your approach. Reflect on your implementation and areas for improvement. Refer to item 6. This may be around 0.5 pages.

8 Template Notes

You can remove this section or comment it out, as it only contains instructions for how to use this template. You may use subsections in your document as you find appropriate.

8.1 Tables and figures

See Figure 3 for an example of a figure and its caption.

8.2 Equations

An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command. This an example cross-reference to Equation 1. You can also write equations inline, like this: $A = \pi r^2$.

Team Contributions

Write in this section a few sentences describing the contributions of each team member. What did each member work on? Refer to item 7.

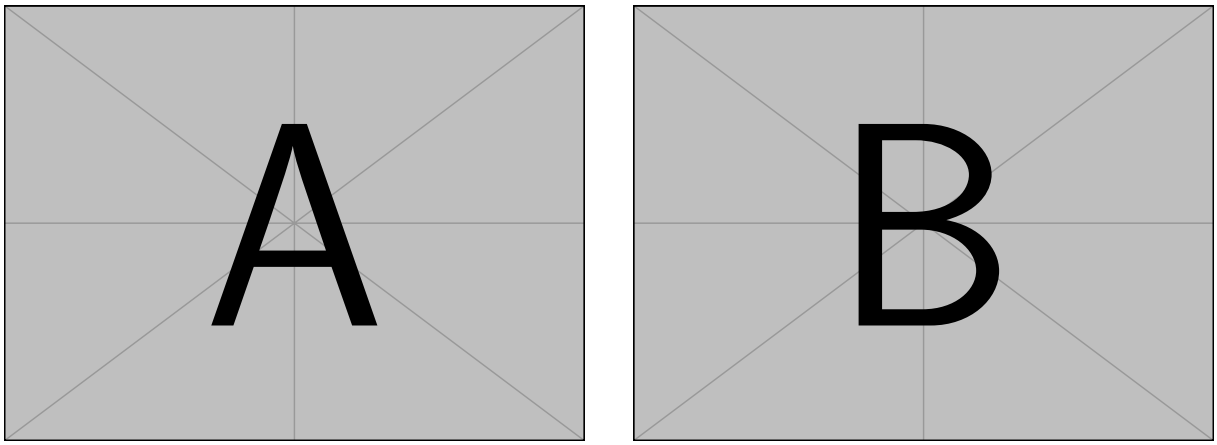


Figure 4: A minimal working example to demonstrate how to place two images side-by-side.